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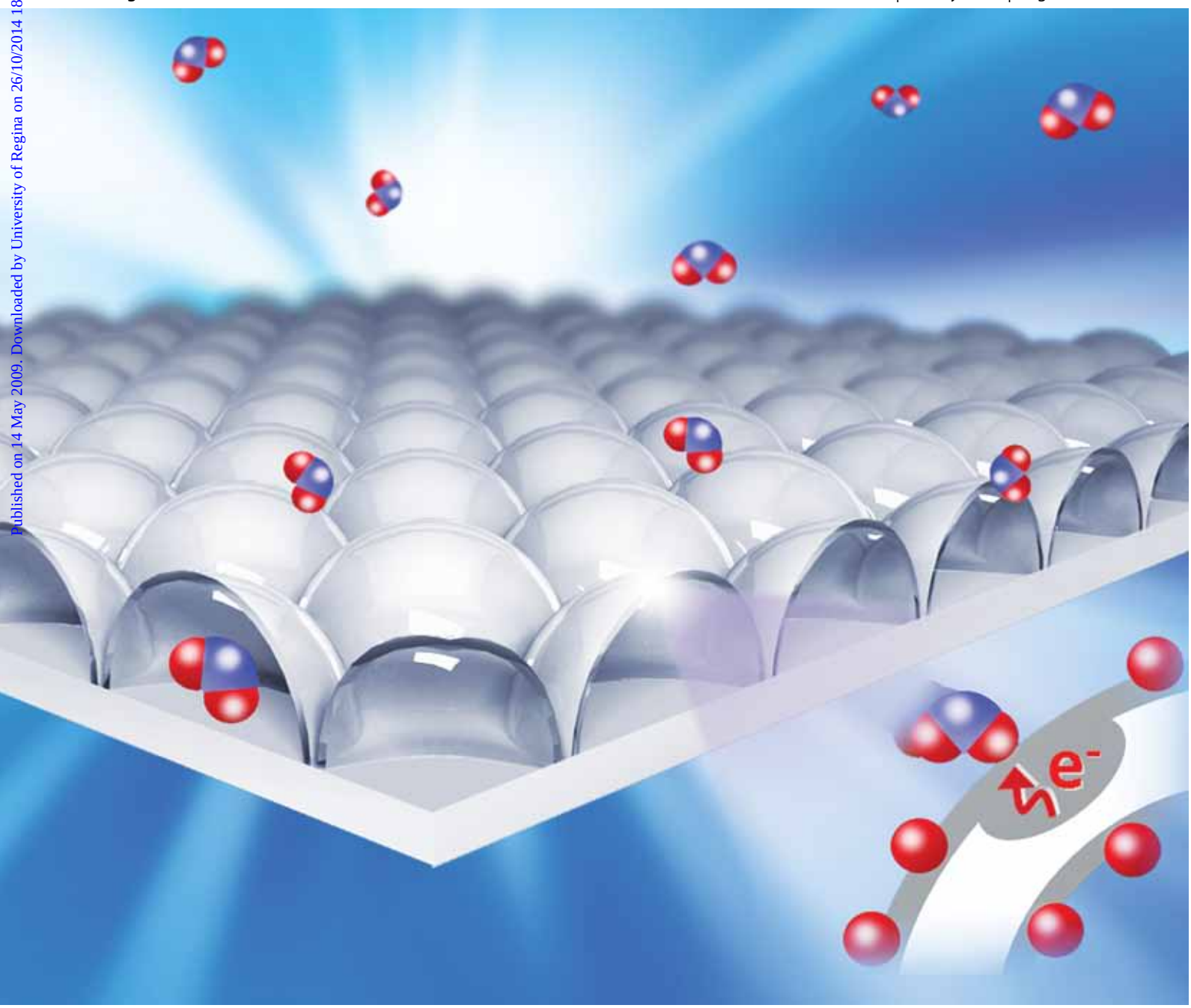

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Fabrication and gas sensing properties of hollow SnO_2 hemispheres

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Close packed arrays of hollow SnO_2 hemispheres were prepared using PMMA microspheres as sacrificial templates for subsequent sputter-deposition of SnO_2 films, leading to a threefold enhancement in gas sensitivity compared to non-templated (flat) films.

Increasing requirements for environmental monitoring, air-quality control, medical diagnosis, and detection of explosive and toxic gases have led to growing interest in ultrasensitive gas sensors. While the detection limit of state-of-the-art solid state gas sensors typically lies in the single ppm (parts per million) range,[1,2] a much higher sensitivity level that enables reliable gas detection at the ppb (parts per billion) level is necessary for some of the advanced applications in these areas. To meet this stringent requirement various nanostructured sensor architectures have been explored aimed at achieving ultra-high sensitivity levels. These include mesoporous layers,[3–6] nanowires,[7–12] nanotubes,[13,14] and nanofibers.[15,16] Despite these efforts, developing versatile and reproducible fabrication methods of ultrasensitive gas sensors remains a challenge. Recently, sacrificial templates comprising close packed silica or polymer microspheres were used to prepare macroporous layers comprising (quasi-)ordered arrays of metal oxide hollow hemispheres.[17–23] These unique "egg shell" structures provide a high surface area while keeping the interface area with the substrate small, necessary conditions to achieve high sensitivity.[24] Furthermore, their fabrication method is simple and versatile involving reproducible and clean physical vapor deposition (PVD) processes. These merits identify them as promising candidates for gas sensors and other applications that require high surface areas such as dye-sensitized solar cells.[25]

In this work we report on the preparation of close packed arrays of hollow SnO_2 hemispheres ("egg shells") and on their gas sensing properties towards NO_2. The preparation method involves sputtering SnO_2 thin films onto poly(methyl methacrylate) (PMMA) colloidal templates on alumina or Si substrates. Following subsequent calcination the organic template decomposes while the inorganic overlayer remains intact, leaving a close packed array of hollow SnO_2 hemispheres. The process can be repeated to produce double-layered structures and so on. The gas sensitivity of these SnO_2 "egg shells" was found to be significantly higher compared with planar SnO_2 thin films deposited under the same conditions on untreated substrates.

PMMA microspheres were used to prepare the colloidal templates. An aqueous suspension (1 g, 2 wt%) of PMMA microspheres (800 nm in diameter) was ultrasonically treated for 30 min and dripped by a pipette onto the substrate. Subsequently the specimens were dried at room temperature in air to obtain the colloidal templates comprising two-dimensional hexagonal close packed (2D-HCP) arrays of self-assembled PMMA microspheres. SnO_2 thin films were deposited by RF sputtering onto the colloidal templated substrates as well as on untreated substrates serving as reference specimens for comparison purposes. The following sputtering conditions were used: RF power of 80 W, working pressure of 10 mTorr, and Ar flow of 5 sccm (5 cm^3 min^{-1}). The deposition time was varied to obtain SnO_2 films of different thicknesses: 100, 400 and 800 nm. The substrates were not heated to prevent thermal decomposition of the PMMA microspheres during film growth. Following SnO_2 film deposition the specimens were calcined at 500 °C for 1 h in air to remove the organic templates and induce crystallization of the SnO_2 films. The phase composition was examined by X-ray diffraction (XRD, RIGAKU, D/MAX-RC, 12 kW). The films were amorphous in the as-deposited state but upon calcination they crystallized to polycrystalline cassiterite (SnO_2 rutile structure) displaying strongest reflections from the (110), (101) and (211) planes, indicating that the SnO_2 hemispheres were untextured.

The morphology and microstructure of the SnO_2 hemispheres (after the final calcination step) were examined by scanning electron microscopy (SEM, PHILIPS, XL30SFEG) and transmission electron microscopy (TEM, FEI, Tecnai F30 S-Twin), respectively. Fig. 1 shows typical SEM images of mono- and double-layered hollow SnO_2 hemispheres on Si and alumina substrates. The SnO_2 hemispheres displayed 2D-HCP structure (*cf.* Fig. 1a), retaining the structure of the PMMA microspheres that results from their self-assembly during drying of the PMMA aqueous suspension. Defects such as vacancies and domain boundaries were observed (*cf.* Fig. 1a) but overall the structure remains quite ordered. During the calcination step the PMMA microspheres decomposed while the inorganic SnO_2 overlayer remained nearly intact, resulting in SnO_2 hemispherical replicates of the PMMA microsphere arrays. The gas pressure that built up inside the SnO_2 hemispheres during

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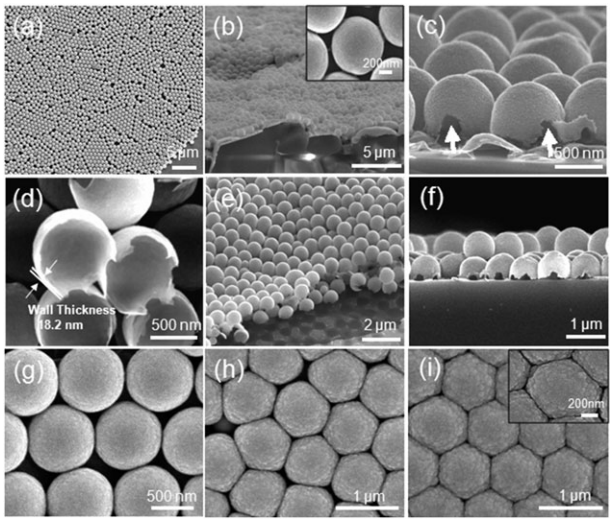

Fig. 1 SEM images of hollow SnO_2 hemispheres: (a) plan view image of a monolayer of 2D-HCP array of SnO_2 hemispheres on a Si substrate; (b) tilted view image of a monolayer of SnO_2 hemispheres on an alumina substrate (the inset shows a high magnification plan view image); (c) cross section image of SnO_2 hemispheres on a Si substrate (burst holes are marked by arrows); (d) inside view of SnO_2 hemispheres that were removed from the substrate and flipped upside down; (e) tilted view image of double-layered SnO_2 hemispheres on a Si substrate; (f) cross section image of double-layered SnO_2 hemispheres on a Si substrate; (g,h,i) plan view images of SnO_2 hemispheres with a nominal wall thickness of 100 nm (g), 400 nm (h), and 800 nm (i). Images (a) through (g) correspond to SnO_2 hemispheres with a nominal wall thickness of 100 nm.

decomposition of the PMMA microspheres was eventually released by rupturing a small segment of the SnO_2 hemispherical overlayer, leaving burst holes at the bottom of the hemispheres (*cf.* Fig. 1c).

The inside morphology of the SnO_2 hemispheres was examined by scratching a piece of the "egg shell" layer and flipping it upside down. The SEM image in Fig. 1d shows no residues of the organic template inside the SnO_2 hemispheres which are completely empty from the inside. The wall thickness of the SnO_2 hemispheres was found to be considerably smaller at the bottom of the layer than at the top. For instance, for a nominal SnO_2 film thickness of 100 nm the wall thickness at the bottom of the hemispheres was approximately 18 nm (*cf.* Fig. 1d). This wall thickness gradient results from shadowing effects. The effect of increasing the nominal wall thickness from 100 to 800 nm is shown in Fig. 1g–i. This was achieved simply by increasing the deposition time of the SnO_2 overlayer. The shape of the SnO_2 hemisphere changed from nearly ideal spheres at small (100 nm) film thickness to distorted polygons (pentagons or hexagons) at intermediate (400 nm) thickness to nearly ideal hexagons at large film thickness (800 nm). Likewise, the packing density increased and the empty spaces (voids) between the spheres/polygons decreased with increasing film thickness, and so did the surface roughness of the SnO_2 hemispheres.

Multi-layered arrays of hollow hemispheres can be produced by repeating the fabrication process (templating, sputtering, calcination) over and over again. An example of a

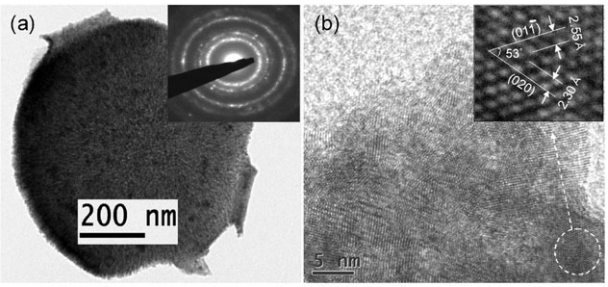

Fig. 2 (a) TEM image of a hollow SnO_2 hemisphere with the SAED pattern in the inset; (b) lattice image from the edge of the SnO_2 shell with lattice fringes from the (01$\bar{1}$) and (020) planes in the inset.

double-layered stack of SnO_2 hemispheres is shown in Fig. 1e and f. Note that the packing density and order is sustained through the upper layer, enabling 3D close packed arrays of hollow hemispheres to be built up layer by layer using this method.[25]

Fig. 2a shows a TEM micrograph of a hollow SnO_2 hemisphere with a nominal wall thickness of 100 nm. The sphere diameter is *ca.* 800 nm, commensurate with the diameter of the PMMA microspheres. The selected area electron diffraction (SAED) pattern in the inset shows that the SnO_2 shell is polycrystalline, in agreement with the XRD results (not shown). The grain size varies between 10 and 15 nm. Fig. 2b shows a lattice image measured from the edge of the SnO_2 shell where the thickness is thin enough to observe lattice fringes. Interplanar distances of 2.55 and 2.30 Å are observed with an angle of 53° between the respective planes (see inset). These planes correspond to the (01$\bar{1}$) and (020) planes of the SnO_2 cassiterite (rutile) crystal structure.

In order to examine whether this microsphere templating approach indeed enhances the gas sensitivity we measured the response of sensor prototypes comprising a monolayer of hollow SnO_2 hemispheres on alumina substrates fitted with interdigitated gold electrodes (16 fingers, 8 mm long and 200 μm wide, spaced 200 μm apart) and compared them with reference specimens comprising SnO_2 films deposited on untreated substrates. The SnO_2 films were deposited under exactly the same conditions in both cases (templated *vs.* non-templated substrates) and were annealed together to insure that the only difference between them arises from the unique morphology obtained by introducing the colloidal templates prior to the SnO_2 deposition step.

The gas sensing measurements were carried out by monitoring changes in the DC current under constant applied voltage (50 mV) during cyclic exposure to trace concentrations (between 125 ppb and 2 ppm) of NO_2 in dry air. To achieve this, a pre-mixed NO_2–air gas mixture containing 50 ppm of NO_2 in dry air (BOC UK) was mixed with clean dry air using mass flow controllers (MKS). To ensure that the residual concentrations of NO_2 and other interfering gases in the reference gas (air) were well below the minimum exposure level (125 ppb) of the test gas (NO_2) we passed fresh air through a zero air generator (Environics Series 7000 zero air generator) comprising a catalytic converter and Purafil and charcoal scrubbers. The zero air generator is specified by the manufacturer to provide dry air (dew point < -10 °C) with

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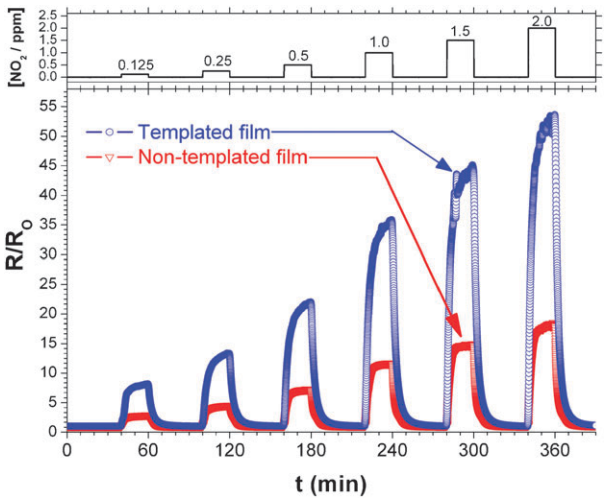

Fig. 3 The resistance response R/R_0 (where R is the resistance at time t during the measurement and R_0 is the baseline resistance in dry air at the beginning of the measurement) during cyclic exposure to increasing NO_2 concentration in dry air at 250 °C. The NO_2 gas concentration profile is shown on top.

residual concentrations of less than 0.5 ppb for NO, NO_2, SO_2, and O_3 and less than 25 ppb for hydrocarbons and CO. The gas sensing measurements were carried out in a small (~0.5 l), cylindrical UHV-compatible stainless steel chamber fitted with gas inlet and outlet ports and a temperature-controlled specimen holder (hot stage). The measurements were carried out in the temperature range 200 to 400 °C taking the necessary measures to ensure a stable and reproducible baseline at the beginning of each test. To this end the specimens were equilibrated under the baseline conditions, that is in clean air, at the test temperature for several hours (typically around 12 h) prior to the first cycle of exposure to the test gas (NO_2).

Representative results of the resistance response of templated (hollow hemispheres) and non-templated (reference) thin film SnO_2 sensors at 250 °C are shown in Fig. 3. Both sensors responded quickly (response and recovery time of a few minutes) and reversibly displayed a significant increase in their resistance upon exposure to NO_2 (typically for SnO_2 gas sensors). The templated films displayed a threefold enhancement in sensitivity compared to their non-templated counterparts. Interestingly, this ratio comes close to the geometric enhancement in their surface area (compared to flat films), which stands at 3.6 for an ideal 2D-HCP array of hemispheres. Other contributions to the enhanced sensitivity may arise from the drastically reduced interface area between the sensing film and the substrate. This interface introduces deleterious effects such as interdiffusion, solid state reaction, and the so-called "dead layer" effect that degrade the sensitivity and long term stability of thin film sensors.[24] By introducing sacrificial templates such as the PMMA microspheres presented in this work most of the sensing layer is shifted away from the substrates and becomes virtually self-supported. As a result, the deleterious effects connected with the film/substrate interface are significantly reduced paving the road towards enhanced sensitivity and more stable performance of thin film gas sensors.

In summary, a hybrid processing strategy that employs PMMA microspheres to prepare sacrificial colloidal templates for subsequent deposition of SnO_2 by means of RF reactive sputtering was described as a means to fabricate close packed arrays of hollow SnO_2 hemispheres. Structural features such as hemisphere diameter, wall thickness and grain size can be tailored by changing the size of the PMMA microspheres and the deposition time of the SnO_2 overlayer. These variations also affect the shape and packing density of the SnO_2 shells. The templated films showed significantly enhanced gas sensitivity compared to conventional (flat) thin films due to higher surface area and smaller interface area between the film and the substrate. The fabrication method described in this work is simple and versatile, providing fascinating opportunities to control the microstructure and morphology of hollow hemispheres of various inorganic materials. In addition to 2D arrays of hollow hemispheres 3D multilayer stacks can also be prepared by repeating the process over and over again.

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